Pre-Calculus 11 - Reasoning and Analyzing Units 1+2 (V1)

Name : _____

Teacher's name: _____

Outcome	0	1	2	3	4	5	6	7
Reasoning and Analyzing								

1. For each of the following statements, identify if the statement is always true, sometimes true, or never true. Provide a brief justification.

   (a) Let a, b, c be non-zero integers, then $ax^2 + bx + c$ can be factored, if there are two integers m, n such that $m \times n = ac$ and $m + n = b$.

   (b) Let A, B be any real number, then $A + B\sqrt{3}$ is an irrational number.

   (c) If $A, B \geq 0$, then $A\sqrt{B} = \sqrt{A^2 B}$.

   (d) If $x^2 + bx + c$ can be factored, then $x^2 + bx + c = 0$ has two solutions for x.

   (e) If $A > 0$, then $\sqrt{A} > \sqrt[3]{A}$.

2. A student was asked to solve the equation and leave answers in exact form. Which step did he first make an error? Explain the error.

$$(2x+1)^2 = 4x - 8$$
$$4x^2 + 4x + 1 = 4x - 8 \quad \textbf{Step 1}$$
$$4x^2 + 9 = 0 \quad \textbf{Step 2}$$
$$(2x+3)(2x-3) = 0 \quad \textbf{Step 3}$$
$$x = \pm\frac{3}{2} \quad \textbf{Step 4}$$

3. In each of the following, one item doesn't belong....

a) Which expression doesn't belong?

i. $\sqrt{8x}$ ii. $\sqrt[3]{8x}$ iii. $\sqrt{18x}$ iv. $\sqrt{12x}$

b) Which equation doesn't belong?

i. $x^2 - 13x + 42 = 0$ ii. $\sqrt{x-6} = x - 6$

iii. $x^2 + 7x + 6 = 0$ iv. $(7-x)(6-x) = 0$